SHA-256

BY

DOUGLAS QUIGG

RESEARCH AND THEORY
WHERE THE STRUCTURE IS, WHERE IT STOPS, AND HOW EACH NULL WAS MEASURED

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ABSTRACT

Measure the null under the same conditions as the effect.

Every claim refuted here, two of them from the published literature, breaks that one rule. Five of them are the same error: the largest of 131,072 dependency cells compared against a threshold built for a single test. The maximum of N standard normal draws sits near $\sqrt{2 \ln N}$ whether anything is there, which for this matrix peaks at 4.85 against a Bonferroni threshold of 5.327. Half a standard error separates the alarm from the null. A fluctuation crosses it now and then and reads as a discovery.

Four results here are analytic. The round is a bijection exactly. All four mixing functions have rank 32 and kernel 0 over GF(2). The narrowed rank deficit is $\deg \gcd(p(x),\ x^w + 1)$, agreeing with elimination at eight widths. And a GF(2) linear variant is pinned at 0.5 at every round by algebra, reading 0.50000000 at rounds 8, 23, 40 and 64. That control detects errors it was not built for.

Reading every cell instead of the largest finds a pattern the maximum cannot see. Taking each cell of the matrix to its class (input bit − output bit) mod 32 puts 4096 cells behind each number. With what every class has in common taken out, the classes that stand are 0, 6, 11, 25 and 31, and each is a named operation: the diagonal, the three rotation amounts of Σ_1, and -1 mod 32 for the carry. The ranking is the same at every depth from round 6 to round 19. What decays at $1/32$ per round is the part every class has in common.

The null is the stronger half. Past round twenty-three the fold reads 2.09, 2.07 and 2.04 against a null peak of 2.63, at three depths, with roughly sixty-four times the sensitivity the maximum has, finding nothing. The wall in the chaining state is the chain length plus two, derived and never searched for, and integer addition is the sole source of algebraic degree while the two nonlinear Boolean functions supply almost none of it.

Everything else here is a measurement, and every measurement states the sample size it was made at.

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1

THE INSTRUMENT

This chapter states the measurement discipline before any result, because every wrong claim about this function fails the same way and the correction is one rule.

1.1 ONE RULE

Measure the null under the same conditions as the effect. Every claim refuted here, two of them from the published literature, breaks that rule in one of three ways.

1.1.1 *A maximum is not a test*

The largest of N draws from a standard normal sits near $\sqrt{2 \ln N}$ whether anything is there. That is a property of taking a maximum and it cannot be tuned away. Compared against a threshold built for a single test, a maximum manufactures signal.

The figure is not small. A 512×256 dependency matrix holds 131,072 cells and its largest deviation peaks at 4.85 standard errors under the null. A Bonferroni threshold at 99% confidence over the same matrix sits at 5.327. The alarm and the null are half a standard error apart. At that separation a fluctuation crosses it now and then and reads as a discovery.

This error appears five times in the material: twice in published work and three times in this one. Each is listed by name in the chapter on refutations.

1.1.2 *A trend is not a signal until it scales*

A statistic built on a sample of size k has a standard error proportional to $k^{-1/2}$. Two things then look identical at one sample size and separate at two:

Noise holds its sigma and shrinks its deviation. Sixteen times the sample leaves the sigma where it was and quarters the deviation.

Signal holds its deviation and grows its sigma. Sixteen times the sample leaves the deviation where it was and quadruples the sigma.

The same test settles both directions. It refutes a published diffusion anomaly whose sigma stays pinned at 4.5 across a sixteenfold range, and it confirms a decay law whose sigma doubles exactly when the sample quadruples.

1.1.3 *A baseline must be measured the same way*

A single timed pass cannot resolve a five percent effect on a machine that is doing anything else. One pass each reads from 0.935 to 1.166 on the same pair of arms, a spread wider than the effect. A first run of any bench on an idle device reads roughly ten percent low while the clock ramps. A warm measurement against a cold baseline therefore invents a gain that is not there.

The throughput figures in this work use best-of-seven with the arms alternating, and the baseline re-measured under the identical protocol.

1.2 AN ANALYTIC NULL

Every null above is statistical: a distribution the measurement is checked against. One better kind is available and is worth more than any of them, because no sample size rescues a wrong one.

Strip SHA-256 to rotations, exclusive-or and constants and it is GF(2)-linear. A one-bit input difference then produces a *deterministic* output difference, the same one for every message. Every cell of the dependency matrix is exactly 0 or exactly 1 and never one half. The deviation is pinned at 0.5 at every round, forever, by algebra and not by measurement.

It reads 0.50000000 at rounds 8, 23, 40 and 64, to eight decimal places.

That control detects errors it was not built for. Removing only integer addition does not make the function linear, because choice and majority use conjunction and stay nonlinear whatever addition does; such an arm reads 0.0093 where 0.5 is required, and fails loudly. The control also refutes the claim that the carry causes the diffusion.

1.3 WHAT FOLLOWS FROM THE DISCIPLINE

Four results in this work are analytic, not statistical, and the argument rests on them:

- The round is a bijection exactly: twenty thousand round trips, zero failures.
- All four mixing functions have rank 32 and kernel 0 over GF(2).
- The narrowed rank deficit is $\deg \gcd(p(x), x^w + 1)$, agreeing with elimination at eight widths.
- The linear variant is pinned at 0.5 at every round.

Everything else here is a measurement, and every measurement states the sample size it was made at.

2

THE CHAIN

This chapter gives the wall an exact law and decomposes it, putting a depth figure within reach of computation instead of search.

2.1 THE WALL IS THE CHAIN LENGTH

A one-bit difference in the chaining state stops being visible to a per-word Hamming statistic at a definite depth. That depth is not a property of the mixing functions. It is the number of words on the shift chain.

The round function generalizes to any even word count: majority and Σ_0 read the low half's head, choice and Σ_1 read the high half's head, and the two halves join the way the standard joins them. At eight words this is SHA-256 exactly and not a model of it. That row is the control.

words	wall	words + 2
8	10	10
10	12	12
12	14	14
14	16	16
16	18	18

Eight words reproduces the wall of 10 measured independently by a separate depth measurement, and the other rows are read against that agreement.

The offset is structure instead of sample size. The same table returns identical at 2^{20}, 2^{24} and 2^{28} pairs, a two-hundred-and-fifty-six-fold range with no drift. A cliff predicts that and a slope does not: a collapse completing inside one round cannot be pushed by adding samples.

2.2 WHY PLUS TWO

Two of the chain's working variables are mixed. Index zero takes $T_1 + T_2$ and index half takes $d + T_1$. Every other index is $\text{state}[i] = \text{state}[i-1]$. A difference moves down the chain unchanged. So whatever the join produced count $-$ 1 rounds earlier is still sitting at the tail of the chain, and the wall should be a mixing depth plus a transport delay instead of one quantity.

Reading each index's own death depth instead of the largest across them shows it. Every index from the e join to the tail dies at exactly $k + 3$:

words	indices from the *e* join	dies at
8	4, 5, 6, 7	7, 8, 9, 10
10	5, 6, 7, 8, 9	8, 9, 10, 11, 12
12	6, 7, 8, 9, 10, 11	9, 10, 11, 12, 13, 14

The last index is count − 1. wall = (count − 1) + 3 = count + 2. The 3 is the round function's mixing depth, how long T_1 and T_2 take to saturate a freshly computed word, and it does not depend on the word count. The wall therefore tracks the count one for one. The −1 is that the chain is count − 1 hops from the join to the tail and not count.

2.3 THE TWO HALVES ARE DIFFERENT CHAINS

The *a* half is not irregular. Both halves are chains of copies below one mixed head, and both follow an exact law; the laws differ in how they depend on the chain length, and comparing a count-dependent offset against a count-independent expectation makes one of them look like noise.

words	*a* half	*e* half	apart
8	index + 6.00	index + 3.00	3.00
10	index + 7.00	index + 3.00	4.00
12	index + 8.00	index + 3.00	5.00
14	index + 9.00	index + 3.00	6.00
16	index + 9.75	index + 3.00	6.75

The *e* half is index + 3 at every count, exactly. The *a* half is index + half + 2, giving six, seven, eight and nine for halves of four, five, six and seven. The 9.75 at sixteen words is one index misreading where the law gives ten.

So the *e* half's saturation depth does not depend on the chain length and the *a* half's does, growing one round per added word pair.

Which half sets the wall. The *a* half's tail is index half − 1 and dies at (half − 1) + half + 2 = count + 1. The *e* half's tail is index count − 1 and dies at count + 2. The *e* half tail is one round deeper. The wall comes from the *e* half even though the *e* half saturates faster per index. It wins by extending to the end of the chain and not by being slower.

Unconfirmed: *a* takes T_1 plus T_2, both halves fully mixed, while *e* takes *d*, a raw copy of the *a* half's tail, plus T_1. Twice the mixing into *a* is consistent with it saturating sooner per index, and *e* reading a raw copy is consistent with its offset not tracking the chain. Neither is measured; only the laws are.

2.4 SIX WORDS IS DEGENERATE

Six words does not wall within thirty rounds. At three words per half, choice reads the entire high half and majority the entire low half. Neither function selects among words any more. It is excluded on that ground.

3

THE OPERATIONS

This chapter separates what each sub-function does, because the category they share is defined by exclusion and hides three different jobs.

3.1 THE BUCKET IS DEFINED BY EXCLUSION

Cryptography's word for the part a linear model cannot hold is *nonlinear*, which is a negative definition. Choice, majority and the carry are filed together because none of them is linear, not because they resemble each other. Measuring them separately shows three unrelated jobs and one operation doing nothing detectable at all.

3.2 ALL FOUR MIXING FUNCTIONS ARE BIJECTIONS

The schedule pair might be expected to be lossy where the state pair is not, since σ_0 and σ_1 each carry a shift and a shift discards bits where a rotation cannot. All four are GF(2)-linear. This needs no sampling: each is a 32×32 matrix and its image is two to its rank.

function	terms	rank	kernel
Σ_0	ROTR2 ⊕ ROTR13 ⊕ ROTR22	32	0
Σ_1	ROTR6 ⊕ ROTR11 ⊕ ROTR25	32	0
σ_0	ROTR7 ⊕ ROTR18 ⊕ SHR3	32	0
σ_1	ROTR17 ⊕ ROTR19 ⊕ SHR10	32	0

Every one is a bijection. Exclusive-or against two rotations restores what the shift drops. The schedule pair collides no earlier than the state pair. Gilbert and Handschuh [2] state this for the Σ pair; the measurement here reproduces it for all four independently.

3.3 THE NARROWED RANK DEFICIT IN CLOSED FORM

Narrowed, they do lose rank, and the deficit is exact. A rotation right by a on a w-bit word is multiplication by x^{w-a} in $\mathrm{GF}(2)[x]/(x^w + 1)$. Three rotations exclusive-ored is multiplication by a three-term polynomial and the kernel is

$$\deg \gcd(p(x),\ x^w + 1)$$

exactly. Checked against the elimination at eight widths, it agrees at every one, including the two nonzero cases: Σ_0 losing four ranks at width six and Σ_1 losing two at width twelve.

Two narrowings are different objects. Narrowing with the raw scaled amounts can make an amount zero and a term the identity, while clamping forces every amount nonzero and the three distinct. The two narrowed SHA-256s are different objects and their rows must not be read against each other. Measured on the clamped object, Σ_0 carries no kernel at any width and the only damaged width is twelve.

The clamping also manufactures structure the standard does not have. Where two amounts scale onto the same value the distinctness rule steps one apart by exactly one, and twenty-two pairs of amounts sit one apart across the narrowed widths against zero at width thirty-two. None of SHA-256's four functions has an adjacent pair; nearly every narrowed one does.

3.4 THREE JOBS, ISOLATED

Removing one nonlinear source at a time cannot discriminate. Full, no-majority and no-choice all read the same algebraic degree, because whichever is removed the others cover for it. **The operations mix, and from outside the bucket looks homogeneous.** Leaving exactly one nonlinear operation standing on the linear spine isolates what it can do alone.

3.4.1 *Algebraic degree*

Degree is proved without sampling. A Boolean function of degree below d has every d-th order derivative identically zero, and that derivative is the exclusive-or of the output over all corners of a d-dimensional input cube. One nonzero fold settles it.

rounds	full	Maj only	Choose only	carry only	linear
2	3	1	1	4	1
3	6	2	1	6	1
4	9	2	2	9	1
5	≥ 18	3	3	≥ 18	1
6	≥ 18	3	3	≥ 18	1
7	≥ 18	5	5	≥ 18	1

The carry alone reproduces the whole function's degree, and at round two exceeds it. Choice alone and majority alone reach degree five by round seven while the full function is past eighteen by round five. Integer addition is the sole source of algebraic degree in SHA-256 and the two nonlinear Boolean functions supply almost none of it.

The ≥ 18 entries are the cube ceiling of the measurement. No measurement produced them. They establish the separation and say nothing about the carry's absolute degree.

3.4.2 *The whole map*

operation	diffusion, whether	diffusion, speed	degree
carry	no	**yes**	**effectively all**
choice	**yes**	some	almost none
majority	no	no	almost none
Σ_0, Σ_1, schedule, K	no	no	no

Removing addition alone still reaches the same diffusion plateau as the full function. The carry is not what makes diffusion happen; the nonlinearity of choice and majority is, and the carry changes how fast and leaves whether alone. That matches the published finding that removing addition costs about two rounds and does not stop diffusion.

3.4.3 *Majority is redundant on four measures*

Majority contributes nothing detectable to the SAC level, nothing to the dampening ratio, nothing to round-to-round diffusion speed, and nothing to algebraic degree. Both published papers also find the majority-removed variant indistinguishable from unmodified. Whatever it is for, nothing measured here can see it, and the linear axis that might carry it is untested for reasons given in the chapter on refutations.

3.5 THE MECHANISM HAS A SHAPE

Choice and majority act inside the bit lattice, where GF(2) describes them. A carry leaves its bit position, travels where GF(2) has no coordinate, and rejoins one position higher. The measurement says that path outside the lattice carries all but a negligible part of the degree.

It is also why a bijection can appear to destroy information while destroying none. The round stays invertible and the difference stays exact; a GF(2)-shaped instrument simply loses the ability to describe it and reports noise. Remove the extradimensional path and nothing ever becomes indescribable. The linear variant, pinned at 0.5 for all sixty-four rounds, shows that.

4

THE SPECTRAL READING

This chapter reads the dependency matrix with every cell and not its largest. That finds a law the maximum cannot see, and then confirms the null harder than the maximum can.

4.1 WHY A MAXIMUM IS THE WRONG READING

Every strict avalanche reading in the literature takes the largest of 131,072 cells and discards the other 131,071. Discarding them pins the peak at $\sqrt{2 \ln N}$ whatever the function does. A structure spread thinly across many cells is invisible to a maximum and plain to a sum.

SHA-256 is built from rotations, and a rotation makes bit j depend on bit $j - r$. If any of that survives into the dependency matrix it appears as a function of (input bit − output bit) mod 32, spread over every word pair and concentrated nowhere. Folding the cells onto that residue puts 4096 of them behind each number instead of one.

The fold is signed and not absolute, deliberately. Taking magnitudes first would fold noise into a positive bias and manufacture a signal in every class.

4.2 WHAT ALL THIRTY-TWO CLASSES SHARE

Two things put a large offset into *every* class at once. The avalanche is incomplete at these depths, and most cells sit far from one half. And the message schedule imposes a light cone: round t consumes W_t. Message word w cannot affect the state before round $w + 1$, and until then flipping any of its bits changes nothing at all, a cell pinned at $-\sqrt{n}$ exactly. Neither of those depends on the output bit. Both spread evenly over all thirty-two residues and concentrate in none of them.

The fold therefore has two components, and the class mean at each depth separates them: a common mode every class shares, and a class pattern measured from that mean.

4.2.1 *The common mode decays at one thirty-second per round*

The loudest class, read without subtracting the class mean, tracks the common mode with a fixed offset. Its amplitude in standard errors over rounds twelve to twenty-three:

rounds	loudest residue	sigmas	reading
12	26	15773.15	common mode
13	26	13724.82	common mode
14	26	11676.61	common mode
15	26	9627.78	common mode
16	26	7580.54	common mode
17	26	5546.15	common mode
18	26	3551.43	common mode
19	26	1916.81	common mode
20	26	789.13	common mode
21	26	160.61	common mode
22	1	13.13	collapsing
23	12	2.09	flat
32	8	2.07	flat
64	31	2.04	flat

The first four differences are 2048.33, 2048.21, 2048.83 and 2047.24. One sigma of the class mean at that sample is 1.526×10^{-5}. 2048 sigmas is exactly $1/32$.

The common mode loses one thirty-second of its amplitude per round, one bit position of thirty-two going diffuse, holding to four decimal places for five rounds, then accelerating into collapse at twenty-two. It is a quantized linear decay and not a smooth exponential, as a function made of discrete operations should produce and as a continuous model would not predict.

Residue twenty-six is not a ridge. Its deviation from the class mean is −1.36 sigmas at rounds eight, twelve and sixteen alike, against a null peak of 2.63 for the largest of thirty-two, and it is a deficit:

round	raw at 26	class mean	excess	scatter	sigmas
8	−23964.8	−22049.6	−1915.2	1410.9	−1.36
12	−15773.8	−13857.8	−1916.0	1411.2	−1.36
16	−7580.8	−5665.9	−1914.9	1411.0	−1.36

A reading that takes the loudest class and never subtracts the class mean reports a fixed ridge at residue twenty-six decaying at $1/32$ per round. The decay is real and belongs to the common mode.

4.2.2 *The law at four times the sample*

The law comes out of a sample where 2048 sigmas happens to equal one thirty-second, and 2048 is 2^{11} while the trials are 2^{18}. Every clean power of two in that arithmetic comes from the chosen sample size, and a law exact at only one sample size is the shape of the measurement and not a law.

	2^{18} trials	2^{20} trials	ratio
round 12	15773.15	31546.74	2.0001
round 13	13724.82	27450.50	
round 14	11676.61	23354.13	
differences	2048.33	4096.24	
	2048.21	4096.37	

Sigma doubles for four times the trials, which is $\sqrt{k}$ exactly, while the absolute deviation holds at $4096 \times 2^{-17} = 1/32$. That is the signature of signal, and the precise opposite of the refuted avalanche dip, which holds its sigma and shrinks its deviation.

4.2.3 *The class pattern does not move*

The fold with the common mode removed is not empty, and what it holds is the round function's own bit transport read straight off the matrix. The ranking is identical at every depth from six to nineteen.

rank	class	sigmas at round 12	what it is
1	0	+3.95	the diagonal: bit j drives bit j
2	25	+1.84	Σ_1, ROTR25
3	6	+1.60	Σ_1, ROTR6
4	31	+1.38	-1 mod 32: the carry, bit k reaching bit $k+1$
5	11	+1.31	Σ_1, ROTR11
6	22	+0.64	Σ_0, ROTR22

Residue zero holds +3.95 sigmas and is constant to four significant figures across eleven rounds, from six to sixteen, while the common mode falls by a factor of nearly three. The scatter of the other classes is constant at 1174 over the same span. The field is two components: a common mode that decays and a class pattern that does not move at all. Nothing rotates, and there is no moving structure for an inverse rotation to unwind.

Classes zero and thirty-one are predictable in advance: any account of this function expects the diagonal and the carry to dominate. **The finding is that Σ_1's three amounts appear and Σ_0's barely do.** All three of $\{6, 11, 25\}$ land in the top six at every depth, about one chance in 225 at a single depth, or one in 112 allowing that Σ_0 would have counted equally.

The mechanism is the asymmetry of the two halves. Σ_1 acts on e and feeds T_1, which is added into both the a chain and the e chain. Σ_0 acts on a and feeds T_2, which reaches the a chain only. Twice the reach, and it is visible in the spectrum.

This also explains why majority measures as contributing nothing on four separate tests. **Majority and Σ_0 are both in T_2**, and T_2 is the weak side. The mixing of SHA-256 is in T_1.

4.3 THE STRONGEST NULL IN THIS WORK

Past round twenty-three the fold reads 2.09, 2.07 and 2.04 against a null peak of 2.63, at three depths, with 4096 cells behind each number and not one. That is roughly sixty-four times the sensitivity the maximum has, finding nothing.

The harmonic sweep agrees. The round-twenty-three profile wanders within two sigmas with no visible pattern, and its discrete transform peaks at f_{13} with magnitude 1.80 against a null peak of 2.35: no repeating trend, no harmonics, no side lobes.

4.4 WHAT THE SPECTRAL READING SETTLES

The common mode decays by a law, one thirty-second per round from round twelve to twenty-one, verified at two sample sizes.

The class pattern is fixed: the diagonal, Σ_1's three amounts and the carry, ranked identically from round six to nineteen.

Both are gone by round twenty-three, at sixty-four times the sensitivity of a maximum.

5

CALIBRATION, AND WHAT IS MOVING

This chapter measures what an instrument can see, by pushing a wave of known amplitude through the same code and fading it until the readout loses it. It then asks the question a magnitude discards: not whether anything is large, but whether anything is *moving*.

5.1 A NULL IS WORTH MORE AS A BOUND

Every null in this work says the same unsatisfying thing: nothing was found. That is a statement about the instrument as much as about the function, and it is worth nothing until the instrument's sensitivity is known. An instrument that can see a bias of one part in a hundred and one that can see one part in a hundred thousand both report "nothing" on the same data, and only the second result is interesting.

The harness is therefore made to carry a plane wave of stated amplitude. A matrix is built that is binomial by construction (the same kernel shape, the same shared tally, the same atomics and the same denominator as the real measurement, with the compression replaced by randomness), and into it a bias of $\pm 2^{-(k+1)}$ is driven, positive or negative according to the sign of a wave at each cell's residue. The bias is built without arithmetic on probabilities: a fair bit or-ed with a rare bit at 2^{-k} lands on $0.5 + 2^{-(k+1)}$ exactly, and and-ed with its complement lands on $0.5 - 2^{-(k+1)}$.

Fading k until the readout loses the wave converts "nothing found" into "nothing above this amplitude".

readout	5σ floor	matched to
chi-square over all cells	1.7×10^{-4}	nothing in particular
residue fold	3.05×10^{-5}	residue-structured shapes

The fold is about six times sharper than the chi-square, and the reason is measured and not argued: the injected wave is residue-structured and the fold is the matched filter for it. That ratio is the value of knowing what shape to look for, stated as a number.

What this excludes. No residue-structured wave with a per-cell bias above $3.05{\times}10^{-5}$ survives past round twenty-three. The published avalanche dip of 0.00272 is eighty-nine times that floor.

5.2 THE INSTRUMENT PROVES ITSELF

A calibration is only as good as the evidence that the readout recovers what goes in. Injected at frequency three across the thirty-two residues, the field's autocorrelation comes back as a cosine of period 10.7, and $32/3 = 10.67$.

lag	1	2	3	4	5	6	7	8
along output	45.29	16.63	−9.47	−36.85	−62.98	−52.68	−26.91	−0.78
along input	44.52	18.06	−9.05	−36.51	−62.95	−52.56	−26.84	0.87

The two axes agree to three digits, and that agreement is the reflection symmetry of the residue coordinate $(i - j) \bmod 32$ reading consistently in both directions, and not a separate finding.

5.3 THE FIELD IS WHITE

A designed function should leave a residual with no length scale at all. A field that is *smooth*, correlated between neighboring cells, would be structure that survived the mixing, and it is the kind of thing an eye reports on noise readily enough that it has to be measured and not looked at.

field	largest |correlation|	verdict
harness alone	2.15	white
SHA-256 round 32	2.40	white
SHA-256 round 64	2.06	white

Sixteen lags per field, eight on each axis. The largest of them peaks near $\sqrt{2 \ln 16} = 2.36$ under a white null whatever the data does. All three sit on it. The field carries no length scale in either direction, in SHA-256 and in a matrix that is binomial by construction alike.

5.4 WHAT THE PROBE CONTRIBUTES

The dependency matrix is never the function alone. It is the function convolved with the probe geometry, and the residue coordinate exists only because input and output bits are indexed by thirty-two bit words. The rotations act on that same lattice. A class pattern in the residue coordinate is therefore an interference between the probe's arrangement and the function's, and not a property of either one, and that bounds what any reading of this shape can be taken to mean.

This is not a defect to be removed. It is the reason the fold works at all: a probe aligned to the function's own lattice renders a thinly spread structure visible. An instrument that is part of its own measurement has to say so.

5.5 WHAT IS MOVING, SEEN OR NOT

Every reading so far looks at one round at a time and asks whether anything in it stands above the noise. A structure too faint to clear that floor in any single round is invisible to all of them

however many rounds are looked at, because the rounds are compared one at a time and then discarded.

But the rounds are not independent samples of an unknown thing. They are a deterministic sequence, and the structure at round r is related to the structure at $r + 1$ by the round function, and that relation is carried in the *phase* of the residue spectrum, which a magnitude discards. A structure standing still holds its phase. One drifting at v residues per round advances it by $2\pi fv/32$ each round. Noise does neither.

Summing the complex spectrum across rounds against a velocity hypothesis therefore adds a real structure as R and noise as $\sqrt{R}$: a gain of $\sqrt{42} = 6.48$, available only to whatever is actually moving at the velocity being tried.

Each round takes its own seed. The structure is deterministic and stays coherent whatever the seed; the sampling noise is made independent and decoheres. One shared seed would keep the noise coherent too and the gain would be nothing.

5.5.1 *The coordinate the readout lives in*

The scan is run in drift $= fv$ and not in velocity. The phase a round advances is $2\pi(fv)r/32$. fv is the natural coordinate and v is a degenerate one: every frequency's cone passes through $v = 0$ together, and away from it the same step in v is a step of f times as much phase.

A uniform grid in v is therefore correct at one frequency and wrong at all the others: at $f = 15$ its step is 1.875 in drift where forty-two rounds resolve 0.195, ten times too coarse, and the loudest SHA-256 cell on such a grid sits at exactly $f = 15$. Scanning uniformly in drift is the same map with the degenerate point resolved: a single grid correct at every frequency at once.

5.5.2 *The result*

	harness alone	walking wave	SHA-256	null peak
magnitude	4.02	**131.24**	3.98	4.66
phase only	4.04	9.15	4.15	4.66
best window	4.07	94.58	4.37	4.66
standing still	2.19	3.50	2.25	—

The positive control is the result worth keeping. The wave is injected at frequency three walking one residue per round, at an amplitude the ladder shows sits below the standing floor. Its standing-still reading is 3.50, *under* the null peak: it cannot be found by any single-round instrument at any sample size those instruments could reach. The scan returns it at 131.24, at exactly frequency three and velocity -1.000. That is a factor of thirty-seven over the standing reading, and all of it comes from the motion.

Against an instrument validated that way, nothing in SHA-256's dependency matrix past round twenty-three is moving, at roughly six and a half times the sensitivity of any standing reading: an effective floor near 4.7×10^{-6}.

5.5.3 *Two pitfalls in reading the scan*

Both are the error class this work meets throughout.

Four maxima are not one maximum. The scan reports four channels and each is itself a maximum over a grid. Comparing any one of them to the peak of a single channel is the max-of-N error one level up. The window scan alone searches 42,570 cells because sliding the start and the width multiplies the grid and does not add to it; the union is 52,299, and the peak that has to be cleared is $\sqrt{2 \ln 52299} = 4.66$, not the 4.12 of a single channel. Read against 4.12, SHA-256's 4.37 would be a detection. Read correctly it is flat, and so is the harness at 4.07.

The phase channel is ceiling-limited. Summing R unit vectors cannot exceed R. The statistic is bounded above by $\sqrt{2R}$ whatever the data does, 9.17 at forty-two rounds. The injected wave reads 9.15, which is saturation and not a measurement. Against a null peak of 4.66 the channel has a factor of two of usable range. Discarding amplitude would need many more rounds than SHA-256 has.

6

SHADOWS OF THE OBJECT

This chapter looks at the whole field instead of at a statistic of it, and lets the picture carry an argument that a table of numbers can get wrong.

6.1 EVERYTHING HERE IS A PROJECTION

The object is three-dimensional (512 input bits by 256 output bits by 64 rounds, 8.4 million cells), and nobody looks at that directly. Every reading in this work is a *shadow*: the residue fold is the field collapsed along one axis, and the chi-square is the same object flattened all the way to a single number. A shadow is cast from one angle, and a second angle is needed to see what the first hides.

That matters more than it sounds, because the projection-slice theorem says a shadow's transform is a slice through the object's transform. Shadows are not impressions of the thing; they are the thing, along a plane. Enough of them reconstruct it.

6.2 THE FIGURE THAT SEPARATES THE COMMON MODE

Here is the residue fold as measured, forty rounds across, thirty-two classes down. Density is magnitude on a log scale, and the two character ramps carry the sign because a printed page has no hue to spare.

```
res | 12345678901234567890123456789012345678901
----+-----------------------------------------
  0 | @@@@@@@@@@%%%%#*:ll..               . ,
  1 | @@@@@@@@@@@@@%%%%#*+-;
  2 | @@@@@@@@@@@@@%%%%##*--
  3 | @@@@@@@@@@@@@%%%%#*+     .   , ,    . ,
  6 | @@@@@@@@@@@@%%%#*+ll  ,
 11 | @@@@@@@@@@@@%%%##*cl        .    ,  .
 25 | @@@@@@@@@@@%%%%#*-ll   ,
 26 | @@@@@@@@@@@@@@%%%##*=. ,
 31 | @@@@@@@@@@@@%%%##*++-
```

Every row is the same row. Taking the largest of these thirty-two and reporting its amplitude in standard errors, without subtracting what all thirty-two share, reports the shared part, and what they share is nearly everything they have.

Two things put that offset into every class at once. The avalanche is incomplete at these depths. And the message schedule imposes a causal boundary: round t consumes W_t. Message word w cannot affect the state before round $w + 1$, and until then flipping any of its bits changes nothing at all. Neither depends on the output bit. Neither can concentrate in any residue.

Now the same field with the class mean removed at each depth.

```
res | 12345678901234567890123456789012345678901234567890
----+----------------------------------------
  0 | KWWWWWWWWWWWWWWWWKxi               ..,
  1 | =**###############**+-;           ..
  2 | =cxxddddddddddddddxx:     .         .
  6 | =dKKWWWWWWWWWWWWKKdi ,   ,  .
 11 | =ddKKKKKKKKKKKKKKKKdi    ,  .    ,. .
 25 | =dKKWWWWWWWWWWWWKKdi, , ,,            ,
 26 | =#%%%%%%%%%%%%%%%%#+. ,
 31 | dKKKWKKKKKKKKKKKKdi=
```

The band that lights up is 0, 6, 11, 25, 31, and every one of those is a named operation: the diagonal, Σ_1's three rotation amounts, and -1 mod 32 for the carry. Residue twenty-six is on the other side, negative, part of the complement the positive band forces.

6.3 THE WATERFALL

The light cone is usually drawn as a staircase, and staircase is the wrong word. Nothing steps. Each message word's influence *falls*, from the dead plateau through a cascade of about seven rounds, and each word begins its fall one round after the last. Sixteen overlapping waterfalls, offset in time.

word	rounds past its own wavefront						
1	65717950	56156848	39901486	23149438	7750636	524352	3713
5	65715565	56038264	39759693	22991641	7598227	517213	3795
10	65710620	56030538	39770014	23005491	7602562	514235	3793
15	65712398	56032235	39694484	22916534	7543738	442264	1969

Aligned to its own wavefront every word falls the same way, to one part in 10^4 at the lip and one part in forty five rounds down. The plateau above the lip is 67,108,608.0 to the last digit, being $256(n - 1)$ at $n = 2^{18}$: not a large measurement but an exact structural constant.

6.3.1 *What the plateau is*

It is tempting to read that constant as *the input cannot reach the output yet.* That reading is wrong, and the statistic says so if it is asked properly.

The projection sums z^2 over the row, and z^2 cannot tell a rate of zero from a rate of one: both give $z = \mp\sqrt{n}$ and $z^2 = n$. The sum is sign-blind. A bit that changes nothing and a bit that flips its outputs *every single time* land on the same plateau.

The evidence is a bit that has no business being there. Bit 511 (the most significant bit of message word fifteen) still sits at the plateau at round seventeen, a round after word fifteen has entered. It cannot be causally dead: flipping the top bit of an addend always flips the top bit of the sum, because the carry out of bit thirty-one is discarded modulo 2^{32}. It flips bit thirty-one of a and of e with probability one. Rate one, $z = +\sqrt{n}$, and the same z^2 as rate zero.

The plateau is where the function is still perfectly predictable, and being outside the causal cone is only the special case where the prediction is *no change*. The most significant bits ride one round longer than the rest of their word for a reason that is arithmetic and not causal: they never generate a carry. Nothing about them is yet data-dependent.

The cliff is where determinism ends instead of where causality begins. The constant, the waterfall, the shift invariance and the diagonal at thirty-two bits per round all stand under that reading; only the name changes, and a wrong name misleads further than a wrong number does.

The register chains are the second waterfall and they are ducts instead of mixers. b, c and d are copies of a delayed one, two and three rounds, with no mixing and no attenuation, and information is carried forward intact. The duct length is therefore what sets where the wall sits.

6.4 ROCKS IN THE RIVER

What the residue band shows is not the operations. It is their *wake*.

The difference propagates; Σ_1, the carry and the diagonal correspondence are obstacles standing in that flow, and each disturbs it by being there. A residue class carries amplitude because an operation upstream keeps moving bits by that distance. The disturbance piles up at that offset and stays there.

That reading is not a figure of speech, because an ablation tests it directly and it is the strongest result in this work. Take the rock out of the river and the wake goes: removing Σ_1 drives its classes from 6.59 to 1.92, and class six does not merely fade but flips sign, from +6.83 to −4.28. Removing integer addition drives the carry class from +5.46 to −0.11, because without addition there are no carries and nothing is transported at $-1 \bmod 32$ at all.

It also explains a null. The Doppler scan of the previous chapter finds nothing moving (drift within 0.09 of zero at all fifteen spatial frequencies), and that is not the absence of structure. **A standing wave around a fixed obstacle does not drift.** The round function is the same at every depth, and the only quantity that does vary with the round, the constant K_t, is measured twice by different means to contribute nothing to diffusion. The rocks do not move, and their wake cannot. Zero Doppler is the mechanism reporting itself.

6.5 WHAT THE PICTURE ESTABLISHES

The shadow reading separates the common mode from the class pattern, extends the constant-decrement law of the common mode two rounds earlier, reproduces the common-mode amplitude

from independent seeds to five significant figures, and, through the schedule ablation, places the knee at round seventeen where the message schedule switches on.

7

A SOLITON, AND THE BOUNDARY IT SHARES

This chapter stops squaring the projection, finds the object that squaring hides, and takes it apart until every number in it is accounted for.

7.1 WHAT SQUARING DESTROYS

Every projection in the preceding chapters sums z^2 along an axis, and z^2 cannot separate a rate of zero from a rate of one: both give $z = \mp\sqrt{n}$ and the same square. The sum is blind to sign. A cell that changes nothing and a cell that flips its output *every single time* land on the same value and are drawn at the same height in the same color.

The obvious repair is worse. Summing signed z makes a row that is half never-flipping and half always-flipping come to zero, the same total a fully mixed row gives, and hides the same distinction one level down.

The three states are therefore counted and not summed. A count of zero makes the cell *never*; a count of every trial makes it *always*; any count between the two makes it *mixed*. Those three are exclusive and cover every case, and nothing cancels.

7.2 SOMETHING THAT DOES NOT MOVE WHILE EVERYTHING ELSE DOES

round	never	always	mixed
6	99,051	**312**	31,709
10	66,294	**312**	64,466
16	17,126	**312**	113,634
17	9,840	248	120,984
21	0	0	131,072

The *always* count holds at exactly 312 for eleven consecutive rounds while *never* falls by a factor of 5.8 and *mixed* climbs by eighty-two thousand around it.

It is not standing still in the field. It is traveling.

round	bits	width	cells	words
6	57–191	135	312	1–5
8	121–255	135	312	3–7
10	185–319	135	312	5–9
12	249–383	135	312	7–11
14	313–447	135	312	9–13
16	377–511	135	312	11–15

A rigid object, one hundred and thirty-five bits wide and three hundred and twelve cells in area, moving at exactly thirty-two bits per round without changing form. That is a soliton: a shape that propagates without dispersing. It is stationary in the co-moving frame, and no reading that averages over rounds can see it. It sits at the same slope of thirty-two that a slant-stack finds statistically at 7.18 sigmas, except that this is exact, with no variance across eleven rounds.

7.3 TAKING IT APART

The projection sums over the output axis. It cannot say which output bits these are. Naming them accounts for every number.

part	count	mechanism
aligned, residue 0	256	integer addition
skewed, residue 25	56	Σ_1's ROTR25, alone

The aligned half is arithmetic. Flipping bit k of an addend always flips bit k of the sum, because carries only run upward. Bit k of a message word drives bit k of T_1 and thence of a and of e, two cells per bit, over four words of thirty-two bits: $4 \times 32 \times 2 = 256$.

The word breakdown closes too. The newest word contributes 64 and is purely aligned; the three behind it contribute 78 each, being 64 aligned and 14 skewed; the oldest contributes 14, skewed only, its aligned cells already mixed away. Four times sixty-four is two hundred and fifty-six, four times fourteen is fifty-six, and the certain cells spread perfectly evenly over the eight state words, thirty-nine each, at every depth.

7.3.1 *Where the explanation fails*

Every one of the fifty-six skewed cells sits at residue twenty-five. None at ROTR6, none at ROTR11, none at any of Σ_0's amounts, at rounds eight, ten and twelve alike.

The natural mechanism does not survive that. If carries destroy certainty at every position above the lowest flipped one, then the lowest of Σ_1's three images should be the survivor, and at source bit twenty that is $k - 11 = 9$, below $k - 25 = 27$. Residue eleven should appear. It never does.

The aligned half is understood and the skewed half is an exact measurement without an explanation. That names the observation the next model has to reproduce.

7.4 THE BOUNDARY THIS SHARES WITH THE PUBLISHED WORK

A local-collision construction is built out of probability-one paths, and this object has neighbors in the published work.

7.4.1 *The deterministic limit is twenty-two rounds*

Sanadhya and Sarkar [6] construct deterministic collisions on reduced SHA-2 and record the limit themselves: "Deterministic constructions of up to 22-round SHA-2 collisions are described using the SS local collision."

Their limit is not arbitrary and its mechanism is a boundary this work finds independently. A local collision spans nine steps, and a deterministic construction requires it to sit entirely inside the sixteen free message words, since those can be solved for directly while every later word is a function of them. It cannot begin after step seven ("obtaining deterministic collisions up to 22 rounds did not require the (single) local collision to extend beyond Step 15"), and beyond that some expanded word necessarily carries a difference and determinism ends.

That is the message expansion switching on, which the schedule ablation locates at round seventeen, bit for bit. **The published deterministic ceiling and the knee measured here are the same structural fact**, approached from construction and from measurement. Round twenty-one, where the last certain cell disappears, is a different and later boundary.

7.4.2 *What the sources say*

On the most significant bit. Sanadhya and Sarkar state that "since the XOR and additive differences coincide for the most significant bit, to achieve higher probability, it is advantageous to ensure that many bits in the differential path are MSBs", because a one-bit difference survives addition "with probability one if the differing bit is the most significant bit, else with probability half".

That is *adjacent to* the aligned half of this object and not the same statement. Their claim is that a difference stays weight one only at the MSB. The claim here is that bit k of an addend always flips bit k of the sum, which holds at every k; what the MSB adds is that no carry leaves it. Both follow from the carry running one way, but they are different facts, and the 256 aligned cells count only the second.

On the message expansion. "The message expansion of SHA-2 does not play any role in first 16 steps", stated near-verbatim, and the local collisions of that work span nine steps.

On the avalanche measurements. Vaughn and Borowczak test all 127 non-empty combinations of the seven sub-functions, and name Σ_1, integer addition, Choose and the message scheduler as those contributing to the avalanche criterion at the earliest rounds throughout. That is the T_1 asymmetry of the operations chapter, found by ablation and not by spectrum, and their result precedes this one.

Their round figures want quoting carefully. Full SHA-256 exhibits the criterion at round twenty-three, *fails it between rounds fifty-four and fifty-seven*, and exhibits it again at fifty-eight. Twenty-three is the floor across combinations, not the full function's plateau.

This work disagrees with them about the dip. The fifty-four to fifty-seven failure does not reproduce at roughly four times their sample, as the refutations chapter records. That disagreement is left standing and not reconciled: their threshold is a Bonferroni bound at ninety-nine per cent over 131,072 simultaneous tests, and the dip sits close enough to it that sample size decides the answer.

8

THE BOUNDARY AS AN INSTRUMENT

This chapter states what a boundary reading can recover, what it provably cannot, and which of the boundary results are algebra instead of measurement. The strongest claim in it is a counting argument that neither precision, sample size nor the number of probes can move.

8.1 THE OCTANT DECOMPOSITION

Split $\mathbb{R}^3$ by the sign of each coordinate. The eight regions tile space with no gap and no overlap, each is an infinite cone with a trilateral right-angle corner, and all eight corners meet at the single point at the origin.

On the boundary the same split cuts eight spherical triangles, and each has *three* right angles. By Girard's theorem a spherical triangle on the unit sphere has area equal to its angle excess,

$$A = \alpha + \beta + \gamma - \pi = \frac{3\pi}{2} - \pi = \frac{\pi}{2},$$

and eight of them come to 4π, the area of the sphere. The tiling is exact and the eight pieces are congruent. No letter of the alphabet below is larger than another by construction. The measured shares total 1.000000000000000.

That gives an eight-letter alphabet: the fraction of the lit set sitting in each octant, one word of eight numbers per state. It is the coarsest reading in this work, and §8.4 is the reason it is presented with its limit attached.

8.2 AN INDEX SHIFT IS A SCREW

Place index k of N on the golden spiral: height $1 - 2(k + \frac{1}{2})/N$, longitude $k\gamma$ with $\gamma = \pi(3 - \sqrt{5})$.

Shift every index by n. Longitude advances by $n\gamma$ and height by $-2n/N$, a rotation about the polar axis composed with a slide along it. That is a screw. Its pitch is

$$\frac{\text{slide}}{\text{turn}} = \frac{-2n/N}{n\gamma} = \frac{-2}{N\gamma},$$

independent of n. One axis and one pitch therefore describe every shift amount there is, and the amounts differ only in how far along the one helix they travel. The move is rigid in height and longitude and keeps area; it is not an isometry of the sphere, because the axial radius $\sqrt{1 - y^2}$ changes under the slide.

Measured at $N = 256$ over six shift amounts:

worst turn error	9.948×10^{-14} rad
worst slide error	0, exactly
pitch	-0.003255258
spread of the pitch over six amounts	4.337×10^{-19}

Indices that run past N wrap and return at the other end, and they return *as a rigid second arm of the same helix*. Wrapping leaves the screw intact and splits it into two arms of one screw. A rotated state therefore reads as a shifted spiral with a second arm laid beside it, never as a scramble.

This is the strongest structural statement in the chapter. The rotation is the most-used operation in the compression function, and on this placement an index shift is one rigid move along one helix whose pitch does not depend on the amount. It is an identity instead of a measurement: the numbers above check the arithmetic, and the pitch spread of 4×10^{-19} is the rounding of the division.

The caution is equally exact. The placement is chosen, and the identity belongs to the golden spiral. A different placement gives a different move. The identity says that SHA-256's rotations *can* be read as one rigid move, never that the function contains one.

8.3 DEFLECTION AND TORSION: AN EXACT INTRINSIC AND CONTEXTUAL SPLIT

A lit point at direction $\hat{n}_i$ and radius r_i contributes to the boundary coefficients

$$a_{lm} \mathrel{+}= q_i \, K_l(r_i) \, Y_{lm}^*(\hat{n}_i), \qquad K_l(r) = (r/R)^l \, e^{-l(l+1)\tau}.$$

The map from sources to coefficients is therefore *linear*, and the field of many sources is a sum that never needs solving. Both kernels are diagonal in the degree: depth reweights degree by degree and so does conduction, and neither mixes one degree into another. That diagonality is Laplace's, and every part of the physics between a source and the boundary reduces to one number per degree because of it.

Now turn the whole lit set about the polar axis by α. Every longitude gains α, and since the coefficients carry $e^{-im\varphi}$,

$$a_{lm} \longmapsto a_{lm} \, e^{-im\alpha}.$$

Two readings of the same coefficients then behave in opposite ways.

Deflection, the power per degree $P_l = \sum_m |a_{lm}|^2$, is invariant. Under a rotation about the polar axis each $|a_{lm}|$ is individually unchanged; under a general rotation the orders mix, but the degree-l subspace carries a unitary irreducible representation of the rotation group, and P_l is invariant under *all* of SO(3). A magnitude cannot see how the object is turned. Rotation blindness here follows from the construction, and no resolution limit enters into it.

Torsion, the phase $\arg a_{lm}$ of the same numbers, moves by exactly $-m\alpha$ under a rotation about the polar axis, sign included. It recovers the angle, and the sign of the recovery is the handedness.

Measured on a rotation of a ring placement:

deflection moved by	2.539×10^{-14}
torsion recovered the angle to	1.803×10^{-13} rad

Where the operations of interest are rotations, the bit positions are carried in the phase, and the power spectrum cannot recover them at any precision. The reading that survives being turned and the reading that measures the turn are complementary, and a caller has to know which of the two carries its signal before reading either.

8.3.1 *The same split in the literature, sampled instead of solved*

Varley, Mediano, Patania and Bongard [8] compare topological data analysis against multivariate information theory on the same data, and separate *intrinsic* higher-order information, invariant to a rotation of the embedding, from *contextual* information, which depends on the embedding. On fMRI data over 36,300 region triads they report normalized O-information correlating with average void persistence at $\rho = -0.65$, $p < 10^{-100}$, and with the number of voids at $\rho = -0.32$; after a PCA rotation those correlations weaken substantially while remaining significant. The contextual part carries much of what they measure.

The split above is the same split with the group action known in closed form instead of sampled. On the sphere the rotation group acts on the coefficients by a character, $e^{-im\alpha}$. Magnitude and phase separate the intrinsic from the contextual *exactly*: the intrinsic half moves by 10^{-14} and the contextual half recovers the move to 10^{-13}. Their caveat applies here too: they state that no formal proofs relate the two frameworks and that their relationships are correlational. Nothing here proves one either. The sphere adds one case where the separation is algebraic and can be checked to the last digit.

Two further points from the same paper bear on this reading. Their synthetic tests find spheres and hollow toroids dominated by the higher-order term of their measure, and the object read throughout this work is a deforming two-sphere boundary. That is a reason to expect higher-order structure in a boundary reading and never a demonstration that there is any. Their distinction also names something visible in the viewer: the shape of the envelope is the intrinsic reading and its orientation is the contextual one.

8.4 WHAT A BOUNDARY CANNOT SAY

A reading to degree L carries exactly $(L+1)^2$ real numbers about its source, whatever the source is. That is how many coefficients there are. The lit set here has 256 degrees of freedom. Where $(L+1)^2 < 256$ the reading is a linear map of rank at most $(L+1)^2$ from a 256-dimensional space. Its null space has dimension at least the shortfall, and every direction in that null space is a change to the object that moves no coefficient at all.

Both kernels sit below one and both are diagonal. Either can only shrink a singular value. Measured on the golden placement at full depth and no conduction, the most favorable case there is:

degree	coefficients	rank	blind	least kept
4	25	25	231	4.495
8	81	81	**175**	4.365
12	169	169	87	3.818
15	256	256	**0**	3.474×10^{-3}
16	289	256	0	1.524×10^{-1}

The map has as much rank as it has rows at every degree below the source count. The shortfall in coefficients is the entire cause of the blindness. Three consequences follow, each a number instead of an opinion.

The viewers read to degree eight, carrying 81 **of** 256**.** Neither sample size, precision nor the number of probes completes that reading. Raising the degree does, and the cost is coefficients instead of accuracy: the basis is orthonormal to $2.2{\times}10^{-14}$ at degree fifteen under exact quadrature. Precision is not what holds the reading at eight.

Degree fifteen is the floor, because $(L+1)^2$ first reaches 256 there. Below it the blindness is forced by counting alone.

Reaching the floor is not reaching a usable reading. At degree fifteen the rank is complete and the least singular value collapses to 3.5×10^{-3}, three decades under the values beside it; at sixteen it recovers to 1.5×10^{-1}. A square map is invertible and an ill-conditioned one is not invertible in any useful sense, and the count alone hides that. One degree of headroom buys a factor of 44 in conditioning.

The eight-letter alphabet is rank 8**, blind in** 248 **of** 256 **directions**, and at fixed weight its shares carry one constraint and leave seven free numbers. This frames the coherence result: sixty-four rounds write sixty-four distinct signatures, which reads as complete separation and comes nearly free, because seven numbers separate sixty-four arbitrary states whether the seven mean anything. Distinctness at that sample size is not evidence. Evidence would be a delta below the redraw level, or a signature that predicts the round it came from.

8.4.1 *The bound arriving as an implementation defect*

A recorded frame that is not carried through the query moves four fifths of the placement and a few percent of the reading. An arm is stated in some frame, and the derivation uses that frame; dropping it between the record and the query is a plain implementation defect. Measured against the eight sign octants over the 256-point golden placement, it sends 199 of 256 points to a different arm. The eight letters move by 3.92 percent in the units the delta is reported in, and the worst single letter moves 1.96×10^{-2}, or three point crossings. Four fifths of the placement relocates and the reading reports a few percent, because the move lives almost entirely in the 248 directions the alphabet cannot see. Here the margin is fourteen decades above the reading's floor and the defect is caught, but the margin is a property of the floor and not of the defect: the letters understate the move by construction, and the count of reassigned points does not.

A letter is a sum, and a sum hides a pair of moves that cancel inside it. Two points exchanging arms leave both letters where they were while the arms are no longer the arms, and a comparison made only at the letters can then report an exact match while the topology moved. The redraw null therefore compares at three levels: each arm's weight at each placement point,

the count of points that changed arm, and the letter a caller reads. The first two are what the claim is about; the third is a consequence.

8.4.2 *The null space is more than truncation*

Truncation is the bound above and the one that binds here. The deeper limit does not go away at unbounded degree: for the wave equation at a fixed frequency there exist sources whose exterior field vanishes identically, and surface measurements cannot uniquely determine a volume source [10]. Recent treatments give spherical-harmonic criteria for when a source fails to radiate, for Maxwell's equations (`arXiv:2402.10407`), on uniqueness, reciprocity and low-signature structures (`arXiv:2509.26304`), and for elastic sources (`arXiv:2508.05401`).

The null space of the reading is not exactly that class, and two different things must be kept apart. A source whose exterior field vanishes identically has every exterior coefficient equal to zero, and no arithmetic reaches it. That is a genuine kernel. A source whose *far* field falls below the reader's floor while its near field does not is a different case, and it is no kernel at all. The reading is taken at a finite radius, where a near field is present by definition. Such a source has coefficients that are small and nonzero. Calling them invisible is a statement about the instrument's floor, and the object still carries them.

The depth kernel prices degrees in digits and does not zero them. The gain at degree l is $(r/R)^l$, which is never zero for $r < R$. A degree is unreadable exactly when its gain lands under the floor, and the last surviving degree is digits$/\log_{10}(R/r)$. At $r/R = 62/100$ the count comes back 77 degrees at sixteen digits, 96 at twenty, 144 at thirty and 264 at fifty-five, against predictions of 77.1, 96.3, 144.5 and 264.9: 4.8168 degrees per digit. The object is built once and never rebuilt, and every column that moves belongs to the format. Over the same sweep the identity floor falls 1.0011 decades per digit and the smallest detectable unmodeled source follows it at 1.0256, against the 0.9861 decades per digit recorded for a rotation null measured at eight precisions [12].

The bound that does not move is the more important half. Rank holds at 64 of 64 at every precision in that sweep. Precision buys the size of what can be read. The number of things read stays where counting puts it: $(L+1)^2$ coefficients against N sources still caps recovery at N, and degrees admitted past that count improve conditioning and add no rank. That is the same statement as degree fifteen being complete at 3.5×10^{-3} while degree sixteen recovers to 1.5×10^{-1}. Arbitrary precision moves this obstruction from principle to price and leaves the counting where it was.

The instrument shows a piece of this directly. The transport deposits into the same coefficient field as the object's constituents, and the two sum, because the reading map is linear and a boundary has no way to report which source put a given part of the field there. Sweeping the coupling gain, the field's span reads 6.993 for the object alone, 6.66 at a low gain, and 14.45 at a gain of six. The dip is no artifact: the beam's coefficients partly *cancel* the object's. Two fields on one boundary add, and adding can subtract. A partial cancellation is a partial nonradiating combination, and it means a beam can hide part of an object as easily as reveal it.

The supported form of "the beams map the whole object" is this: **beams map the radiating part, exactly, and the null space not at all, where the null space is taken at the precision of the reading, and that boundary moves.** Coherence has to be defined on the quotient by that null space, or it counts invisible differences as differences. The quotient is therefore indexed

by precision. A coherence figure is a statement about a reading in a format, and two figures taken at different widths are not comparable without saying so. The part that does not move with the format is the nonradiating class above, and the quotient by that class is the same at every width.

8.5 DEFORMATION WITHOUT MOTION

The claim under test is that the boundary contorts while nothing moves: the far side of the sphere stays undisturbed, the enclosed volume stays stable, no coordinate is displaced, and therefore in the topological sense there is no motion, only deformation.

The coefficients support a precise version of this. The reading is a section over a *fixed* sphere, and a deformation is a change of the coefficient vector at fixed support. Degree zero is the monopole, the integral over the whole sphere, and it holds the conserved quantity: to the extent a_{00} holds, the reading redistributes instead of adding.

The measured effect is a large amplification of a small move. Over one full run the degree-zero swing reads 0.12 while the degrees above it swing 0.27 to 0.88. The low degrees sit nearly fixed and the high degrees cascade, and a small move at the bottom accompanies a large move everywhere else.

The attribution matters more than the measurement. Both kernels are diagonal in degree. Depth and conduction reweight degrees and never mix them. A degree-zero plateau is therefore stable under both, and either kernel most strongly amplifies or kills high-degree structure. Conduction alone is a low-pass at degree about $1/\sqrt{\tau}$. The amplification belongs to the kernels, and it must appear for any object read this way, which is also why the measurement is not evidence about SHA-256 by itself.

Two limits belong with it. The even and odd lobe split and the exponential falloff visible in the power bars are artifacts of the ring placement and the depth control, never of the hash. And the degree-zero swing above is read off the viewer over one run: an instrument reading with no error bar.

8.6 THE DELTA, AND THE OBJECT IT IS READ ON

The eight-letter word, round by round, on the working state of the compression function, against a floor measured the same way from two unrelated states of the same weight, redrawn:

	delta	against the floor
first eight rounds	11.9%	1.27×
last rounds	7.1%	0.76×
redraw floor, unrelated states	9.4%	1.00×

The early rounds move the word *further* than independence does, and the late rounds move it less; the delta crosses the floor between the two ends of the run. A rigid relabeling of the indices carries mass from octant to octant without mixing it, which puts the delta above the floor, and mixing puts it below. Attributing the early excess to the register shift is a reading of why, and is not measured.

The finalized digest is the wrong object. The finalization adds the chaining input back, modulo 2^{32}, and that destroys exactly the structure being looked for. Counting the six words each round copies from another:

carried words matching, on the working state	378 of 378
carried words matching, on the finalized digest	0 of 378

On the digest the register shift is not present at all, and a measurement on the digest that cannot see the shift reads as the alphabet saturating. It does not saturate. The instrument chapter's rule is to measure the null under the same conditions as the effect; this failure sits one level beneath it. **Name the object being read, because reading the wrong one fails silently and in the direction that looks like a result.** A quantity with a trend also must not be assigned a level, whether a ramp, a stationary value or a floor; the refutations chapter lists that shape.

8.7 THE TRANSFORM TABLE

Three sections above end at the same observation from different directions: both kernels are diagonal in the degree, and that diagonality is Laplace's. The observation generalizes, and the general form is short enough to tabulate and strong enough to price an instrument with.

A reading is the coefficient vector a of length $(L+1)^2$. Every operation in this work is an operator on that vector. Writing diag for the property of acting on each degree by a single number, without moving power between degrees:

transform	action on a_{lm}	diag	invariant
rotation, general	$\sum_{m'} D^{l}_{m'm}(R)\, a_{lm'}$	block	P_l
rotation about the read axis	$a_{lm}\, e^{-im\alpha}$	yes	P_l, each $\lvert a_{lm}\rvert$
depth, outward carry	$(r/R)^l\, a_{lm}$	yes	direction per degree
conduction, smoothing	$e^{-l(l+1)\tau}\, a_{lm}$	yes	direction per degree
pixel prefilter	$e^{-l(l+1)\sigma^2/2}\, a_{lm}$	yes	direction per degree
band truncation	keep $l \le L$	yes	every kept coefficient
point inversion	$(-1)^l\, a_{lm}$	yes	P_l
index screw	rotation $n\gamma$, slide $-2n/N$	no	area
synthesis	$f(\hat{n}) = \sum a_{lm} Y_{lm}(\hat{n})$	no	none
translation of the origin	degree l draws on degrees 0 to l	**no**	degree 0
octant indicators	project onto eight indicators	**no**	total weight

One theorem accounts for the diag **column.** The Laplace–Beltrami operator on the sphere has the degree-l subspace as its eigenspace,

$$\Delta Y_{lm} = -l(l+1)\, Y_{lm},$$

and any operator commuting with Δ preserves its eigenspaces and therefore acts on each degree separately. Every diagonal row commutes with Δ, for a reason particular to it. Depth is harmonic

continuation: the field satisfies $\Delta f = 0$ in the shell, and separating variables leaves $(r/R)^l$, the exponent being the degree because the degree indexes the eigenvalue. Conduction and the pixel prefilter are both $e^{\tau\Delta}$, the heat semigroup, and an operator built as a function of Δ commutes with it by construction. Rotation and point inversion are isometries of the sphere and Δ is built from the metric; a mirror through the equatorial plane gives $(-1)^{l+m}$ instead. Truncation is a spectral projection.

The exceptions are instructive. The index screw is not an isometry: its slide takes $Y_{10} \propto y$ to $y - 2n/N$, which has a degree 0 part, and only its rotation is diagonal. A translation is not a map of the sphere to itself. There is no Δ on the sphere for it to commute with, and re-expanding a field about a shifted origin is triangular in degree, degree l of the new expansion drawing on degrees 0 to l of the old. Every cheap operation in this work is diagonal and the operation it avoids is not. The instrument gives each source its own depth factor instead of moving an origin, and the table shows why that choice is forced.

8.7.1 *The composition law*

Diagonal operators on shared eigenspaces commute, and composing them multiplies one number per degree. The chain from an interior state to a drawn pixel collapses to a vector of $L + 1$ gains,

$$a_{lm}(\text{drawn}) = g_l \, a_{lm}(\text{state}), \qquad g_l = (r/R)^l \, e^{-l(l+1)\tau_{\text{total}}} \, [\, l \le L \,],$$

with the smoothing times *adding*:

$$e^{-l(l+1)\tau_1} e^{-l(l+1)\tau_2} = e^{-l(l+1)(\tau_1+\tau_2)}, \qquad \tau_{\text{total}} = \tau_{\text{conduction}} + \tau_{\text{pixel}}.$$

At the ceiling the instrument runs to, g is eleven numbers, recomputed when the depth, the smoothing or the ceiling moves. Where the pixel width varies per pixel the gains are per pixel, at one exponential and $3L - 2$ multiplications each.

8.7.2 *A pixel asks for an average, and a band-limited field has it in closed form*

A pixel does not ask for the field at a point. It asks for the field averaged over the solid angle it covers, and every antialiasing scheme is an attempt at that average. Supersampling estimates it: k samples cost k times as much and converge about as $1/k$, reaching the answer only in the limit.

Convolution on the sphere is multiplication in the degree. Averaging against a zonal kernel K is therefore $a_{lm} \mapsto K_l a_{lm}$, with no sampling anywhere, and choosing K to be the heat kernel of angular width σ gives the prefilter row of the table with $\tau_{\text{pixel}} = \sigma^2/2$. **One evaluation of the prefiltered field is a Gaussian-weighted average over the pixel's footprint, exactly.** The exact average over a cap is also diagonal; a square footprint is not zonal and is not diagonal. The comparison with supersampling is therefore not between two approximations: supersampling approximates a quantity this field gives in closed form, and by the composition law that factor is an addend in a τ already being applied.

The same exponential says which degrees may be skipped. $e^{-l(l+1)\tau}$ falls under a threshold ε once $l \gtrsim \sqrt{2\ln(1/\varepsilon)}\,/\,\sigma$, about $3.33/\sigma$ at one level of an eight-bit channel. **This saving is small.** It reaches down to degree ten only when σ is about 0.33 radians, an object about six pixels across;

above that size every coefficient contributes something visible. The level-of-detail behavior is derived instead of authored, and it earns nothing at working sizes.

8.7.3 *What the instrument draws instead, measured*

The viewer samples the 121 coefficients at 2,701 vertices of a 36 by 72 grid, five degrees to a cell, and the rasterizer fills each cell with two planes. The picture on screen is a straight line drawn through a curve the coefficients hold exactly. Measured against exact evaluation, as a fraction of the picture's own contrast, and as an angle between the drawn surface normal and the true one:

field	largest	typical	normal, median	normal, 95th
degree 1 alone	0.0010	0.0004	1.25°	5.85°
degree 10 alone	0.0592	0.0108	7.47°	**47.15°**
flat spectrum to 10	0.0312	0.0050	5.85°	—
63 deposits at $r/R = 0.80$	0.0273	0.0061	6.67°	—
63 deposits at $r/R = 0.40$	0.0075	0.0015	2.14°	—

The value error rises 59 times from degree one to degree ten, an exponent of 1.77 against the 2 a linear interpolant's estimate $\frac{1}{2}(\pi h/\lambda)^2$ gives for a degree-l harmonic of length $360/l$ degrees. Halving the step divides the error by 3.95 against a predicted 4, confirming that the error comes from linear interpolation and no other effect.

The normal columns carry the finding. The surface is shaded and shading reads the gradient, a linear interpolant has a constant gradient inside a triangle, and the drawn normal is a staircase across a field whose gradient turns smoothly. At degree ten the median normal is off by 7.5° and one sample in twenty by more than 47°. That is a far larger error than the one percent in the value column, and it is the five-degree quilt visible on the reconstruction.

The depth kernel is visible in the last two rows. A deeper reading carries less high-degree power, by the depth row of the table, and the same grid draws it four times more accurately. The grid is wrong in proportion to the fine structure the state happens to hold, and never by a fixed amount.

Two limits belong with this. The prefilter is exact for an isotropic footprint, and a pixel's footprint near the silhouette is stretched, leaving a single σ to over-blur across the short axis; the exact repair wants a directional kernel, and that kernel is not diagonal in the degree and therefore leaves the composition law behind. And the prefilter is exact for a band-limited field: the boundary field is band-limited by construction, since a reading is a truncated expansion, while the room's corners and the arms are not, and prefiltering those would ring.

8.7.4 *Every constant here is decidable*

The transforms are built from π, $\sqrt{2}$, $\sqrt{5}$, the golden angle $\pi(3-\sqrt{5})$, and the harmonic unit $1/(2\sqrt{\pi})$. All are computable to any length by integer arithmetic, with the verification carried by published bits instead of by a trusted digit string: SHA-256's sixty-four round constants are the leading thirty-two fractional bits of the cube roots of the first sixty-four primes, and its eight starting

words the same for the square roots of the first eight. Recomputing them checks $72 \times 32 = 2{,}304$ independently published bits. All agree, and all fifteen copies of the table in the source match the definition. The one number this produces: the double the placement runs on sits 4.441×10^{-16} radians from the true golden angle, about 2.8×10^{-14} accumulated over sixty-three indices.

8.7.5 *What a measured floor belongs to*

The calibrated null floor for one reading is 4.005×10^{-16}, a bit above the epsilon of a double. It is either a fact about the boundary reading or a fact about float64, and the two answers lead to different work. The question is settled without a statistical model, because the law supplies the answer in advance.

The predicted residual is exactly zero. Take the null the floor is built on: a ring placement turned by whole ring steps, read for P_l. Each point slides along its own ring. The turned configuration is the same set of directions relabelled, each coefficient pair turns rigidly by $m\alpha$, and the sum of their squares cannot move. Whatever a program reports above zero is its own arithmetic.

That makes the floor measurable in the one way that identifies what it belongs to. Compute the same null at rising precision and watch the residual. A residual falling one decade per decimal digit is the number format; a residual that flattens is an operation in the reading failing to commute with the rotation.

Eight readings span float32 through decimal at sixty-two places: host and device float32 and float64 on an RTX 3070, and four decimal points with pi computed to the working precision by the integer Machin series.

	measured	predicted
slope, decades per digit	0.9861	1.0000
worst departure from the line	0.246 decades	0
span covered	55 orders of magnitude	—

The prediction carries no fitted quantity: a format holding d digits rounds at 10^{-d}. The slope is one before any measurement is taken. Eight points land on that line across fifty-five orders of magnitude, and the largest departure is a quarter of a decade.

The floor is float64's and not the reading's. It remains the correct threshold to hand a caller running in float64, since that is the arithmetic such a caller has. It is wrong as a statement about the boundary: the reading's own residual there is zero, and its floor is whatever precision is bought.

The two arms separate the device from the format. Host against device gives a ratio of 0.906 at float32 and 0.949 at float64. The device's arithmetic is the host's at each width and the accelerator contributes no error of its own. One arm alone would confound the two causes.

Guard digits are arithmetic too. A decimal ladder carrying twelve guard digits, asked for twenty places, rounds at thirty-two. Plotting its residuals against the requested precision instead of the working precision puts a false segment at the join between the float arms and the decimal ladder, at 3.82 decades per digit, and moves the fit to 1.3188 with a worst departure of 6.839 decades. The fit's own departure statistic refuses that reading.

No measured constant appears anywhere in the reading path. The contrast is with the 2022 CODATA adjustment [11], which lists each fundamental physical constant with a value and an uncertainty. About half read *exact*, because the 2019 redefinition of the SI fixed the speed of light, the Planck constant, the elementary charge, the Boltzmann constant and the Avogadro constant by definition; the rest carry an uncertainty no amount of computing will shrink. The reading carries no experimental uncertainty at all, and its only floors are chosen: the degree ceiling, which §8.4 prices; the number format; and the calibrated null, which measures this work's own arithmetic and not the world.

8.8 WHERE THIS SITS

Offered as a placement and not as a result.

The eight octant shares are a partition of the boundary, and a partition is an observable. The category of observables ordered by refinement is an *information structure*, and in that setting the classical information functions appear as cocycles: Baudot and Bennequin [1] construct information cohomology and show Shannon entropy [7] to be the unique measurable nontrivial solution in degree one; Vigneaux [9] extends the setting, and for continuous vector-valued observables finds the only new degree-one cocycles to be differential entropy and the dimension of the support (`arXiv:2107.04377`).

Every reading in this chapter is a functional of a probability vector on *one* partition. All of it lives at a single object of that category. The cohomology is in the maps *between* refinements, where the mutual informations at all orders sit. The alphabet is one coarse-graining and a finer one refines it. The well-posed next measurement is not a sharper delta on the octants but the comparison across the octant, sub-octant and pair readings, a published frame for a measurement not made here.

Every reading should ship a known-null move and report its residual. Every threshold is then measured and never guessed. The coherence figure rounds at 10^{-6} where integer counts over an integer weight have a true null near 10^{-16}: ten decades coarse.

8.9 UNTESTED HERE

The reading above degree eight. The rank argument puts the floor at degree fifteen and usable conditioning at sixteen. Neither is run on the hash.

Whether the numbers a reading carries are the interesting ones. Rank is a ceiling on what can be recovered and says nothing about what is worth recovering. A rank-81 reading blind to every bit of consequence is arithmetically identical to one that is not.

Coherence on the quotient. It is computed on the raw signature, counting invisible directions as though they were visible.

The packing spread of 1.03 **to** 1.10, which carries no error bar at all.

Shape independence, holding only at dimensions three to five and untested outside them.

Any formal relation between the split measured here and the topological and information-theoretic results it is placed beside. The analogy is structural and the published relations are correlational, by their authors' own statement.

9

CLAIMS THAT DO NOT HOLD, WITH WHAT REFUTES EACH

A refuted claim with its cause attached is the cheapest kind of knowledge. This chapter lists the claims about SHA-256 that the measurements refute, each with the measurement that refutes it.

9.1 CLAIMS ABOUT THIS WORK'S OWN MEASUREMENTS

Forward converges, the inverse does not. It does not hold. Read as a statistic and not as a threshold crossing, the inverse scales as $\sqrt{N}$ at every depth from nine to fourteen: ratios of 4.00, 4.00, 3.99, 3.99, 4.22, 3.83 for sixteenfold more pairs. Both directions have walls; only the shape of the approach differs.

The inverse has no wall, sixteen and climbing. It does not hold. It dies at fifteen and round sixteen reads 2.438, 1.317, 0.881 as pairs rise, which is noise shrinking. One more round per fourfold increase is the decay rate being measured instead of a trend to project.

The word statistic goes blind at round ten. It does not hold. The per-bit refinement sees more where there is more (inverse round fourteen reads 18.5 by bit against 7.5 by word), but no depth in either direction reads by bit and not by word.

Forward round ten is small instead of off. It does not hold. At 2^{30} pairs a continued slope predicts 9.706 and it reads 2.386, unmoved across 2^{26}, 2^{28}, 2^{30}. The same extrapolation on the inverse as a control lands within four significant figures at all three sizes.

Decay is geometric, 29.7× **forward against** 2.1× **inverse.** It does not hold. Sweeping from round one shows both accelerate; that comparison puts forward at its steepest against the inverse four rounds earlier on the same shape. A slope of 1.469 per round fitted through rounds pinned at the statistic's ceiling is the ceiling.

The peak destruction rate tracks word width. It does not hold. Peaks read 3.50, 2.05, 1.67, 3.54, 5.24, 3.03, 4.86 bits against widths of 2.00 to 5.00, and the column is not measurable at that resolution regardless.

The schedule pair is lossy because it carries shifts. It does not hold. All four functions have rank thirty-two and kernel zero.

The sparse words step by four. It does not hold. The exact trace floors the nonce at $2, 1, 6, 2, 4, 2, 7, 6, 5, 2, 11$ across W_{18} to W_{28}; only W_{19} is a genuine path and the floor-two words are $18, 21, 23, 27$, spaced three, two, four. The step is which cancellations clear the sampling floor.

A minimum weight per word measures sparsity. It does not hold. A minimum is a tail statistic: at 2^{27} pairs chance alone gives a one-bit word once per word. The rate against $33/2^{32}$ is the measurement.

A floor of one means the path is reachable. It does not hold. W_0 floors at one on W_{33} and is never observed in 2^{25} pairs. The floor wants a carry-free chain and that chance falls with its length.

Nonce bit thirty-one carries 180 **sigma.** It does not hold. A sweep over 2^{31} nonces leaves bit thirty-one zero in every sample, against an expectation computed over the full space. The sweep has to cover the whole 2^{32}.

The 5.0% **word-round saving is worth** 5.0% **of runtime.** It does not hold on the CPU. Implemented, exactly correct on two hundred thousand nonces, it measures 0.998, 0.978, 1.079, 0.957, with a median about 1.00. Word-rounds bound arithmetic, and on the CPU arithmetic does not bound runtime. On the device, where the kernel is ALU and power bound, the same change is worth 4.34%.

A single timed pass can resolve a five percent effect. It does not hold. One pass each reads 0.935 to 1.166, a spread wider than the effect.

Register pressure limits the device kernel. It does not hold. Thirty-eight registers, zero spill stores, zero spill loads, and occupancy capped by the thread limit and not by registers.

Single-bit masks can see linear structure. It does not hold, by its own control. A GF(2)-linear function's output bit is an exclusive-or of many input bits and is balanced against any single one. The known-linear arm reads the noise floor exactly like every other arm. The linear axis remains untested.

The width sweep measures annihilated state functions at several widths. It does not hold. That is the raw-amount object; in the clamped object Σ_0 is clean at every width and only width twelve carries a kernel.

The octant alphabet saturates. It does not hold, refuted by the object and not by the statistic. A measurement on the finalized digest, where 0 of 378 carried words match the words they are copied from, cannot see the register shift at any sample size; on the round's own working state 378 of 378 match. The finalization adds the chaining input back modulo 2^{32} and destroys the structure being looked for. On the state the delta starts at 1.27× the redraw floor and crosses to 0.76×: a trend, with no saturation in it.

The octant delta sits at a level. It does not hold, for any level: a ramp, a stationary value and a floor all fail. The quantity has a trend and crosses its own null between the ends of the run. A threshold placed on it is a threshold on the wrong shape.

Sixty-four distinct signatures out of sixty-four rounds is coherence. It does not hold, by a rank. The eight-letter reading is a linear map of rank 8 on a 256-dimensional source, leaving seven free numbers at fixed weight and 248 directions that move no letter at all. Seven numbers separate sixty-four arbitrary states whether the seven carry meaning. Distinctness at that sample size is free.

A ridge at residue twenty-six decays at $1/32$ **per round.** It does not hold. Residue twenty-six sits -1.36 sigmas from the class mean, a deficit, and the $1/32$ decay belongs to the common mode every class shares.

9.2 CLAIMS TAKEN FROM THE PUBLISHED LITERATURE

SHA-256 fails the strict avalanche criterion at rounds fifty-four to fifty-seven. It does not hold. Measured at three sample sizes, the absolute deviation halves each time k quadruples, $1/\sqrt{k}$

exactly, while the sigma holds at 4.5 across a sixteenfold range. The threshold crossed is 5.327 sigmas against a null that peaks at 4.85 unaided.

SHA-256 never meets the strict avalanche criterion at any round. It does not hold. The persistent 0.21% is 4.2 sigmas at a million trials, *below* what the maximum of 131,072 null draws produces without any function at all.

Choice and majority dampen each other, ratio 0.508**.** Reproduced at 0.5082 and explained. Choice alone gives a SAC-mean of 0.01562361, majority alone 0.01560888, and the two together 0.01562336, identical to choice alone to two parts in ten million. The pair does not dampen; majority contributes no diffusion that choice does not already provide. The measured value is A while the independence model predicts about $2A$, and the ratio is one half by construction.

9.3 THE SHAPE THEY SHARE

Five of these are one error: a maximum over N cells compared against a threshold built for a single test, two published and three in this work. Two more are a trend read from a threshold crossing instead of from the statistic. Two more are a baseline measured under different conditions from the effect. Three more are a level asserted for a quantity that has a trend, the same error inverted.

Two are a different shape and a worse one. **The saturation claim reads the wrong object**, and the coherence claim **counts a separation that a rank makes free**. Neither can be caught by a better null, because the null is fine: one measures a state that does not contain the effect, and the other measures a quantity whose dimension is too small to carry it. The discipline therefore needs a second rule beside the first (**name the object, and count the dimensions of the reading**), and both failures stay silent in the direction that looks like a result.

Refuting them takes no new cryptanalysis. All but two require measuring the null the same way as the effect; those two require asking what is being read.

10

AGAINST THE FIELD

This chapter places the work beside the published record, because assuming a gap is the error a literature check exists to prevent.

10.1 THE FRONTIER IS THREE TIMES DEEPER THAN ANY PROBE HERE

	rounds of 64
practical collision, ASIACRYPT 2024 [4]	**31**
semi-free-start collision, Li and others [3]	**39**
earlier collisions, Sanadhya and Sarkar 2008 [6]	23, 24
this work's forward wall	**10**
this work's deepest probe of any kind	≈ 23

That is not a contradiction. It is two different measurements. This work asks how far a *random* difference stays statistically visible under undirected probing. The published attacks construct a *chosen* differential path, using local collisions and message freedom to cancel differences as they arise. A constructed path survives where a random one is long gone.

No null here means "nothing survives past round twenty-three." It means undirected statistics stop seeing there, and something directed reaches thirty-one in practice. The same caveat applies to every refutation here: undirected probing throughout.

The gap is not sensitivity. The spectral fold is sixty-four times as sensitive as the maximum and stops at the same round. The difference is watching against steering.

10.2 WHERE THE AGREEMENTS FALL

The avalanche onset reproduces exactly. At 2^{20} trials, matching the published million, the SAC value reads 899.7 sigmas at round twenty-one, 152.7 at twenty-two and 4.5 at twenty-three. Round twenty-three, independently, with an independent implementation. That reproduction validates the instrument before it is used on anything contested.

The Σ functions are injective in GF(2), which Gilbert and Handschuh [2] show by the polynomial argument and which the rank measurement reproduces here for all four functions instead of two. The closed form for the narrowed deficit, $\deg \gcd(p(x), x^w + 1)$, extends it; no narrowed-width formula appears in the literature searched.

Majority is redundant. Both published avalanche papers find the majority-removed variant indistinguishable from unmodified, and four independent measurements here agree.

Diffusion contributors. The published work names Σ_1, integer addition, choice and the message scheduler as contributing at the earliest rounds, which is consistent with the mixing depth of three measured independently here.

10.3 WHERE THE EMPHASIS DIFFERS

Both avalanche papers group choice, Σ_1 and integer addition together as diffusion contributors. That grouping is correct for diffusion and wrong for degree. The isolation measurements here say **choice supplies diffusion and the carry supplies degree**, and those are different jobs sharing one category.

10.4 CARRY SPACE IS MAPPED, NOT EMPTY

The carry looks like unexplored ground and is the most heavily mapped part of the subject. Lipmaa and Moriai [5] give exact algorithms for the differential probability of addition modulo 2^n, in closed form and logarithmic time, reducing the hardest case from $\Omega(2^{4n})$ to $\Theta(\log n)$. Additive differential probabilities of exclusive-or, UNAF, bit-vector differential models and work on the non-independence of chained modular additions extend it, with results still appearing in 2025. The thirty-one-step collision is built from that machinery.

The finding that the carry supplies the algebraic degree is therefore a rediscovery of why the field works in carry space instead of a gap in it. The rules there are solved and hard instead of unmapped.

10.5 WHAT APPEARS TO BE UNSTATED

Offered as candidates, not claims, since a literature search cannot prove absence:

- The separation of diffusion from degree across the same three operations.
- The chain law wall $=$ count $+ 2$ and its decomposition into a mixing depth of three and a transport of count $- 1$.
- The two half-chain offsets, one constant in the chain length and one growing with it.
- The closed form for the narrowed rank deficit.
- The quantized $1/32$ per-round decay of the common mode of the residue fold.

10.6 UNTESTED HERE

Constructed differential paths. Everything here probes and nothing builds. The thirty-one-step result is a designed path. Detection is the weaker instrument by a factor of three in rounds, and that gap is the summary of this work's reach.

Message freedom. A collision attack varies the whole message to steer the path. Mining fixes all but thirty-two bits. Most published technique may not transfer, but that is an argument about applicability and not a measurement.

The linear axis, which needs Walsh masks, not single bits, and is a transform over 2^{512} of them.

Majority's purpose, which remains unexplained on every measure available here.

BIBLIOGRAPHY

[1] P. Baudot, D. Bennequin. *The Homological Nature of Entropy*. Entropy 17, 3253-3318. 2015.

[2] H. Gilbert, H. Handschuh. *Security Analysis of SHA-256 and Sisters*. Selected Areas in Cryptography 2003, Lecture Notes in Computer Science 3006, 175-193. 2004.

[3] Yingxin Li, Fukang Liu, Gaoli Wang. *New Records in Collision Attacks on SHA-2*. IACR ePrint 2024/349.

[4] Y. Li, F. Liu, G. Wang, X. Dong, S. Sun. *The first practical collision for 31-step SHA-256*. Advances in Cryptology, ASIACRYPT 2024, doi:10.1007/978-981-96-0941-3_8.

[5] H. Lipmaa, S. Moriai. *Efficient Algorithms for Computing Differential Properties of Addition*. FSE 2001, Lecture Notes in Computer Science, 336-350, doi:10.1007/3-540-45473-x_28.

[6] S. K. Sanadhya, P. Sarkar. *New Collision Attacks against Up to 24-Step SHA-2*. Lecture Notes in Computer Science, INDOCRYPT 2008 91-103, doi:10.1007/978-3-540-89754-5_8.

[7] C. E. Shannon. *A Mathematical Theory of Communication*. Bell System Technical Journal 27, 379-423 and 623-656, July and October 1948.

[8] T. Varley, P. Mediano, A. Patania, J. Bongard. arXiv:2504.10140. 2025.

[9] J. P. Vigneaux. *Information structures and their cohomology*. Theory and Applications of Categories 35(38); arXiv:1709.07807. 2020.

[10] N. Bleistein, J. K. Cohen. *Nonuniqueness in the inverse source problem in acoustics and electromagnetics*. Journal of Mathematical Physics 18(2), 194-201. 1977.

[11] Peter J. Mohr, David B. Newell, Barry N. Taylor, Eite Tiesinga. *CODATA recommended values of the fundamental physical constants: 2022*. Reviews of Modern Physics 97, 025002, doi:10.1103/RevModPhys.97.025002. 2025.

[12] D. Quigg. *Workbook*. orior research papers, https://github.com/dstroy0/orior. 2026.